



INTERHOSPITAL CONFERENCE

**A 10-month-old boy presented with
painless right periorbital mass for 6 months**

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Identification Data

- ผู้ป่วยเด็กชายชาวพม่า อายุ 10 เดือน
- ภูมิลำเนา: ประเทศพม่า
- แหล่งข้อมูล: มารดา (ได้จากการซักประวัติผ่านล่าม) และใบส่งตัว

Chief Complaint

ก่อนที่ตาขวาบวม 6 เดือนก่อนมาโรงพยาบาล

History and Present illness

- 6 เดือนก่อนมาโรงพยาบาล มารดาให้ประวัติมี **ก้อนที่เปลือกตาขวา บวมแดงเล็กน้อย** ลืมตาได้ปกติ ไม่มีไข้ ไม่มีไอ น้ำมูก ไม่มีผื่น
- 3 เดือนก่อนมาโรงพยาบาล **ก้อนที่ตาขวาบวมแดงมากขึ้น ใบหน้าบริเวณแก้มขวา และหน้าผากบวมแดงมากขึ้น จนไม่สามารถลืมตาขวาได้** ไม่มีอาการงอแงมากขึ้น
- มารดาจึงพาไปตรวจที่โรงพยาบาลจังหวัด **ได้นอนโรงพยาบาลและวินิจฉัยเป็น orbital cellulitis** ได้รับการรักษาด้วย **IV clindamycin และ IV ceftazidime ทั้งหมด 10 วัน** อาการไม่ดีขึ้น หลังจากนั้นได้รับการรักษาด้วย **IV methylprednisolone ทั้งหมด 5 วัน** เปลือกตาขวาบวมลดลง เริ่มลืมตาได้ และได้เปลี่ยนเป็น prednisolone 1 mg/kg/day กลับไปรับประทานต่อเป็นเวลา 2 สัปดาห์

History and Present illness

- หลังกลับไปที่บ้าน ก่อนที่ตาขวา แก้มขวา หน้าผาก กลับมาบวมมากขึ้น นอกจากนี้เปลือกตาล่างซ้าย เริ่มบวมแดงเช่นกัน จนไม่สามารถลืมตาได้ทั้งสองข้าง
- 3 วันก่อนมาโรงพยาบาล มีอาการไอมีเสมหะ มีน้ำมูก มีไข้ต่ำ ๆ ไม่ได้วัดไข้ ไม่มีหายใจเหนื่อย
- มารดาจึงพากลับไปตรวจที่โรงพยาบาลจังหวัดแห่งเดิม ได้นอนโรงพยาบาล และรักษาด้วย IV clindamycin เป็นเวลา 10 วัน แต่อาการไม่ดีขึ้น ก่อนที่ตาขวา หน้าผาก และเปลือกตาซ้ายยังบวมแดง พอเดิม จึงได้ส่งตัวผู้ป่วยมารักษาต่อที่โรงพยาบาลมหาราชนครเชียงใหม่

Past History

- เป็นบุตรคนเดียว คลอดที่ค่ายอพยพ ไม่มีภาวะแทรกซ้อน กลับบ้านพร้อมมารดา
- ได้รับวัคซีนครบตาม EPI program ไม่ได้วัคซีนเสริม
- ปฏิเสธโรคประจำตัว หรือยาที่รับประทานประจำ
- ก่อนหน้านี้สุขภาพแข็งแรงดี
- ไม่มีประวัติโรคมะเร็งหรือโรคทางพันธุกรรมในครอบครัว

Physical Examinations at CMU

- **Vital signs:** BT 38°C, HR 127/min, RR 35/min, BP 100/78 mmHg, SpO2 98% (RA)
- **GA:** a Burmese young infant, alert and active
- **HEENT:** no pale conjunctiva, no icteric sclera, no injected pharynx, multiple palpable cervical lymphadenopathies, measuring up to 1 cm in diameter
- **Lungs:** clear and equal breath sound both lungs
- **Heart:** normal S1 & S2, regular rhythm, no murmur
- **Abdomen:** no distension, soft, no point of tender, no hepatosplenomegaly
- **Extremities:** no joint swelling/erythema/warm, CRT<2 secs, pulse full 2+, no deformity
- **Neuro exam:** E4V5M6, good consciousness, pupil 3 mm RTLBE, normotonia, spontaneous equal movement, DTR +2 all extremities

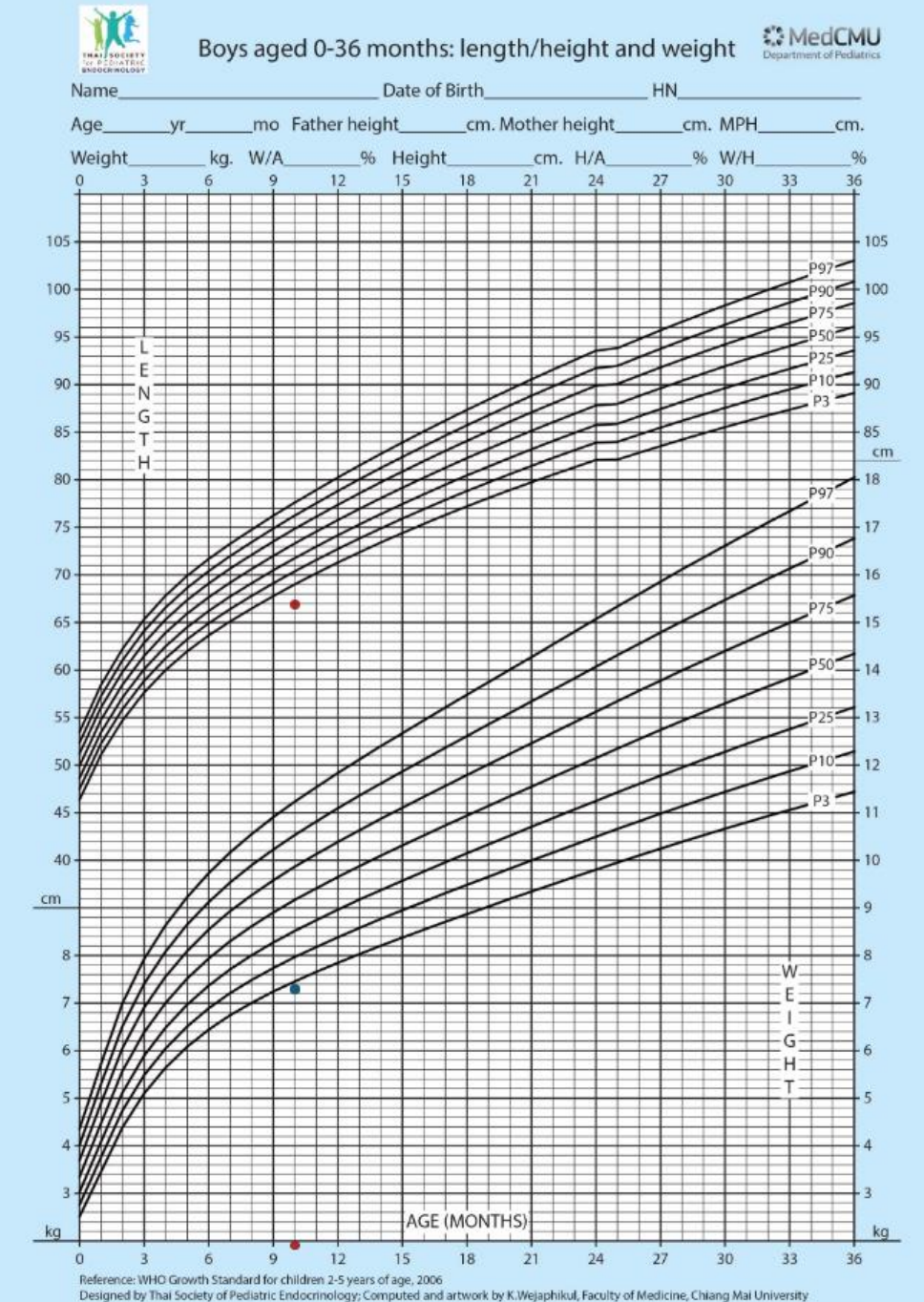
Physical Examinations at CMU



- **Skin:** warmth and diffuse erythema in the right periorbital area, accompanied by an ill-defined, firm to hard, lobulated subcutaneous soft tissue mass. The swelling extends to the nose, the center of the forehead, the infraorbital area of the left eye, and the upper parts of both cheeks. No ulceration over the affected regions.
- **Eye exam:** cannot open right eye, left eye no chemosis, no proptosis, no injection, no discharge

Growth Curve

- BW 7.6 Kg (<P3)
- Length 67.5 cm (<P3)
- W/A 82.6%
- W/H 98.7%
- H/A 91.1%





Problem Lists and Differential Diagnosis ?

Problem Lists

A 10-month-old Burmese infant with

- Slowly progressive painless firm rhinofacial subcutaneous nodule for 6 months with history of treatment with antibiotics (ceftazidime and clindamycin) and steroid
- Acute fever with URI symptoms
- Failure to thrive

Differential Diagnosis

Infection	■ Fungus • Entomophthoromycosis ■ Mycobacterium • Tuberculosis • Nontuberculous mycobacteriosis • Leprosy
Malignancy/Tumor	• Soft tissue sarcoma • Lymphoma
Genetic	• Infantile hyaline fibromatosis

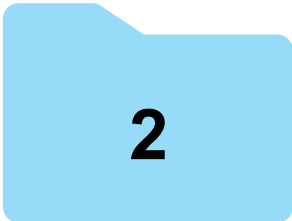


Investigations ?

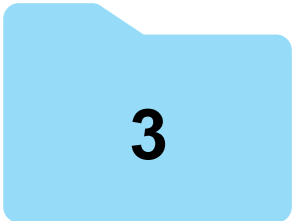
Investigations



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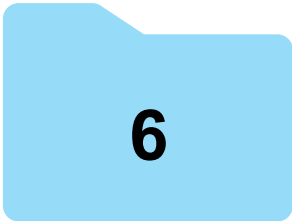
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8

CBC

CBC	
Hb (g/dL)	7.9
Hct (%)	26.5
WBC (cell/cu.mm)	22,220
Neutrophil (%)	40.7
Lymphocyte (%)	45.6
Monocyte (%)	4.4
Eosinophil (%)	9.2
Basophil (%)	0.1
ANC	9,060
AEC	1,999
Platelet (cell/cu.mm)	366,000
MCV (fl)	63.9



Blood Chemistry

Blood Chemistry	
BUN (mg/dL)	2
Creatinine (mg/dL)	0.24
Na (mmol/L)	133
K (mmol/L)	3.9
Cl (mmol/L)	98
CO2 (mmol/L)	18



LFT

LFT	
Total protein (g/dL)	8.5
Albumin (g/dL)	4.2
Globulin (g/dL)	4.3
Cholesterol (mg/dL)	219
AST (U/L) (21-44)	30
ALT (U/L) (9-25)	7
ALP (U/L) (23-98)	175
Total bilirubin (mg/dL)	0.34
Direct bilirubin (mg/dL)	0.18
LDH (U/L)	198



Inflammatory Markers

CRP (mg/L)	72.5
ESR (mm/hr)	43

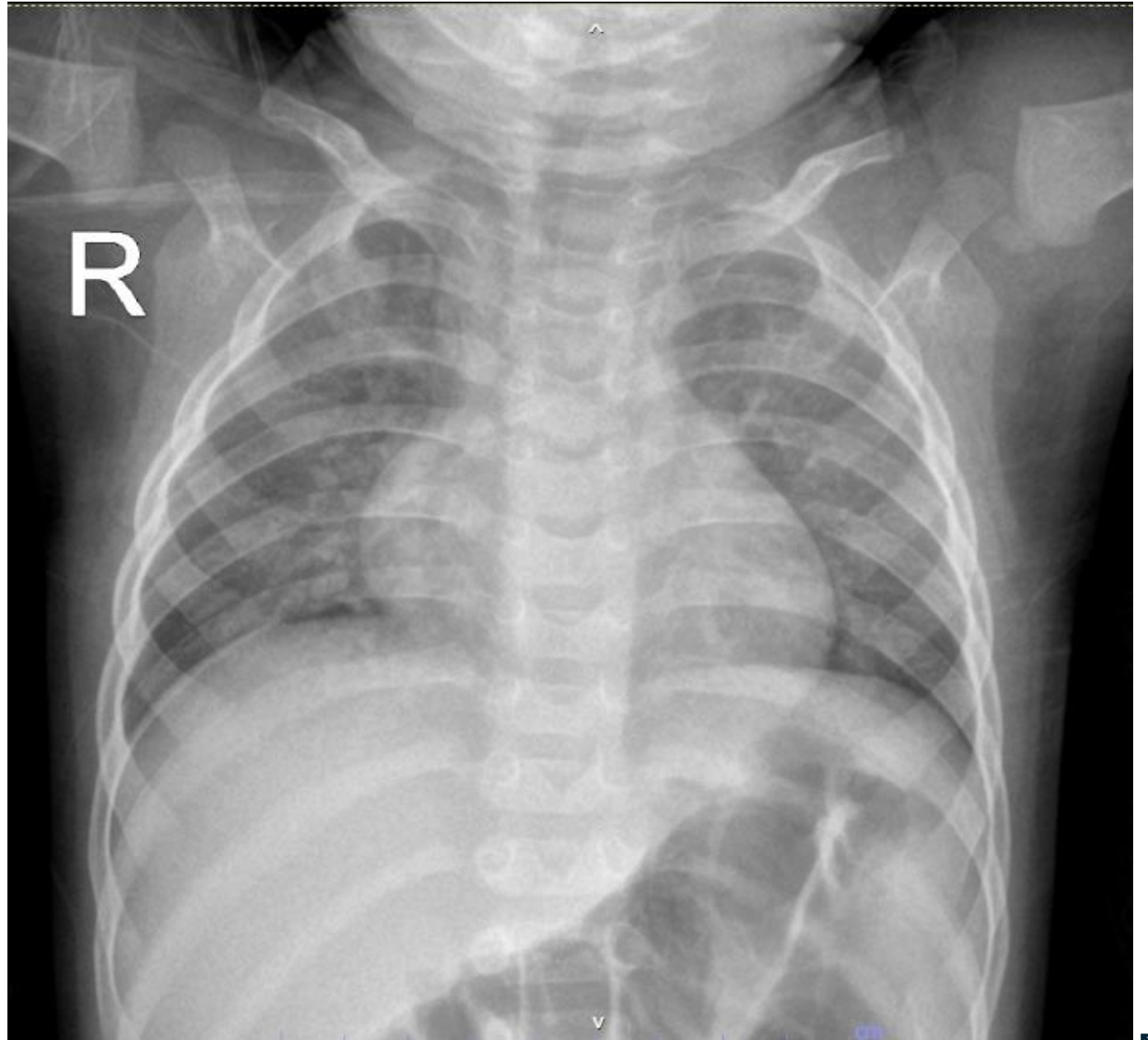


Hemoculture

Hemoculture	
Peripheral	No growth
Peripheral	No growth



Chest X ray



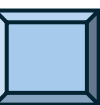
CT orbit emergency



CT orbit emergency

Findings:

- No radio-opaque intraocular foreign body in both orbits. Normal globes and orbits.
- A large infiltrative enhancing hyperdense lesion along skin and subcutaneous layer of bilateral frontal, preseptal, premalar, zygomatic areas, right cheek and right temporal regions could be an infiltrative tumor (e.g. lymphoma) or chronic infection, please correlate with clinical context.
- Multiple enlarged and subcentimeter enhancing lymph nodes along both sides of neck including bilateral jugular, posterior cervical and intraparotid regions, measuring up to 1.2 cm in short axis at bilateral cervical
- The included brain parenchyma and dural venous sinuses appear unremarkable. No gross vascular compromise including intraluminal filling defects or abnormal dilatation. Total opacification of all PNSs, Bilateral mastoid air cells are unremarkable, Intact bony calvaria, base of skull and facial bones



Tissue Biopsy

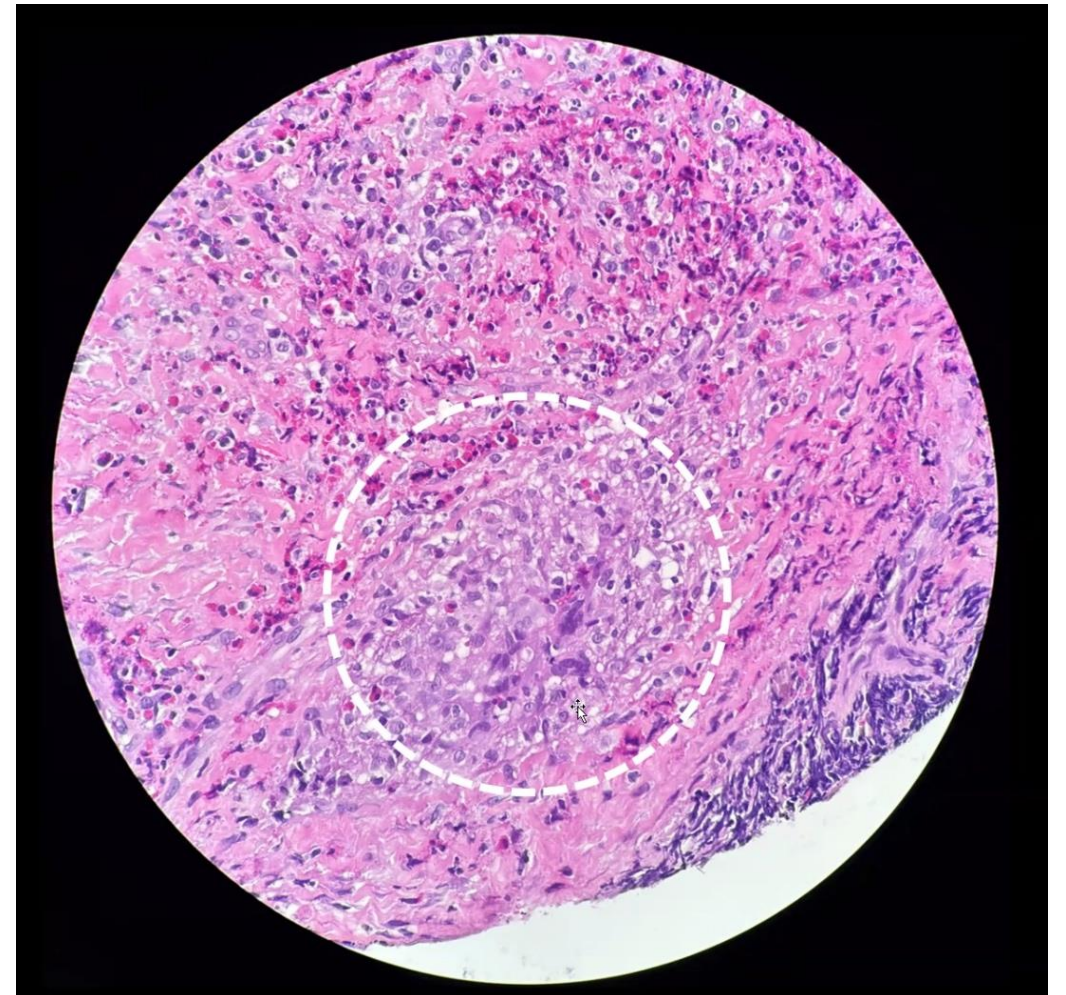
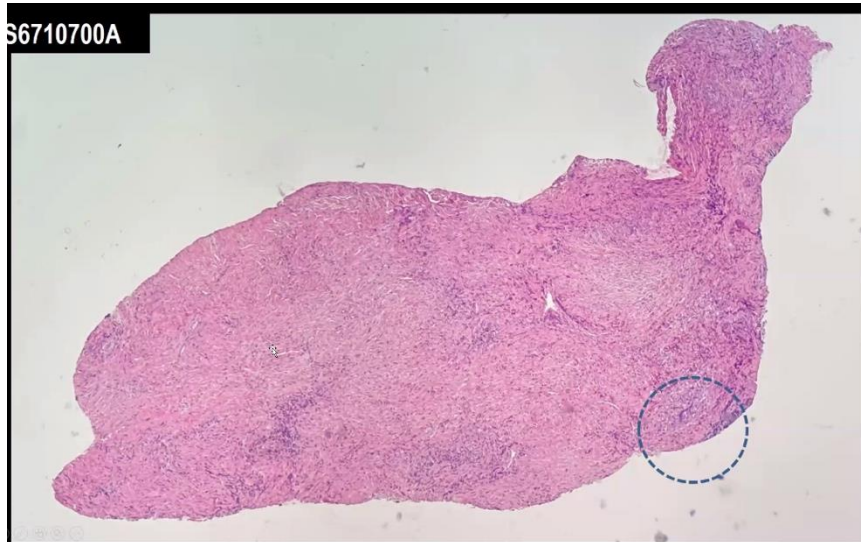
Soft tissue from right eye	
Tissue gram stain	No bacteria
Tissue culture for bacteria	No growth
Tissue Wright's stain	No fungus seen
Tissue culture for fungus	No growth in 2 weeks
18s rRNA	Negative



Tissue Histopathology from Right Eye

Pathology report:

- Fibrosis with many eosinophilic infiltration
- Single focus of non-caseating granulomatous inflammation
- GMS, PAS, AFS, mAFS: Negative



Bone Marrow Aspiration and Biopsy

Findings

- Hypocellular bone marrow (40% cellularity) with few immature B-cells, favor hematogone
- Normal maturation of remaining hematopoietic cells
- Normal number of megakaryocyte with dysplastic-like feature
- No abnormal lymphoid cell infiltration

Consult hematology

- **Impression:** bone marrow suppression from chronic infection

Provisional Diagnosis



Most likely rhinofacial entomophthoramyiasis

Supporting reasons:

- Clinical presentation: chronic progressive facial swelling
- Epidemiological factors: endemic in Southeast Asia
- Host: immunocompetent
- Risk factors: malnutrition
- Histopathology: granulomatous inflammation with eosinophil infiltration
- Imaging: soft tissue swelling. No bony destruction.



Management ?

Progression

DOA 8

1 week after
itraconazole treatment

DOA 23

Management :

- Empiric cloxacillin 100 MKDay and clindamycin 40 MKDay
- Empiric Itraconazole 100 mg/day
- Consult allergy for work up primary immunodeficiency
 - No T-cell& B-cell defect
 - DHR: normal



หลังได้ itraconazole ยังมี facial swelling, redness มากขึ้น

Management:

- Add SSKI solution 300mg/day
- Consult endocrine: suggest monitor TFT

Progression



**2 weeks after
itraconazole treatment**



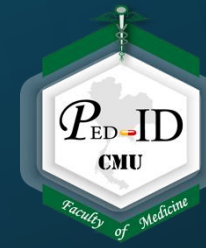
10 days after start SSKI



14 days after start SSKI



21 days after start SSKI



Entomophthoramycosis

Entomophthoramycosis

- Entomophthoramycosis refers to infections caused by members of the order *Entomophthorales*
- The word itself translates in the Greek language to “insect destroyer”, describes the pathogenicity of these fungi in insects
- Two genera are pathogenic to humans and they include *Conidiobolus* spp. and *Basidiobolus* spp.
- Entomophthoramycosis is usually, but not always, limited to the skin and subcutaneous tissues in immunocompetent patients
- Cutaneous and subcutaneous entomophthoramycosis in immunocompetent hosts is largely limited to countries within tropical and subtropical climate zones

Pathogenesis

- The organisms are extremely low virulence in human hosts as manifested by rare occurrence despite ubiquitous nature of fungi
- It has been proposed that infected patients may acquire the organism during a state of transient immunosuppression, perhaps during a viral infection or a state of inadequate nutrition, or as the result of repeated exposure
- Members of *Entomophthorales* commonly found in feces of reptiles, amphibians and other animals as well as in decaying vegetable matter
- The mode of acquisition hypothesized to be through minor trauma, inhalation or ingestion of environmental spores

Clinical manifestations

Chronic rhinofacial
entomophthoramycosis

Chronic subcutaneous
entomophthoramycosis

Gastrointestinal
basidiobolomycosis

Chronic Rhinofacial Entomophthoramycosis



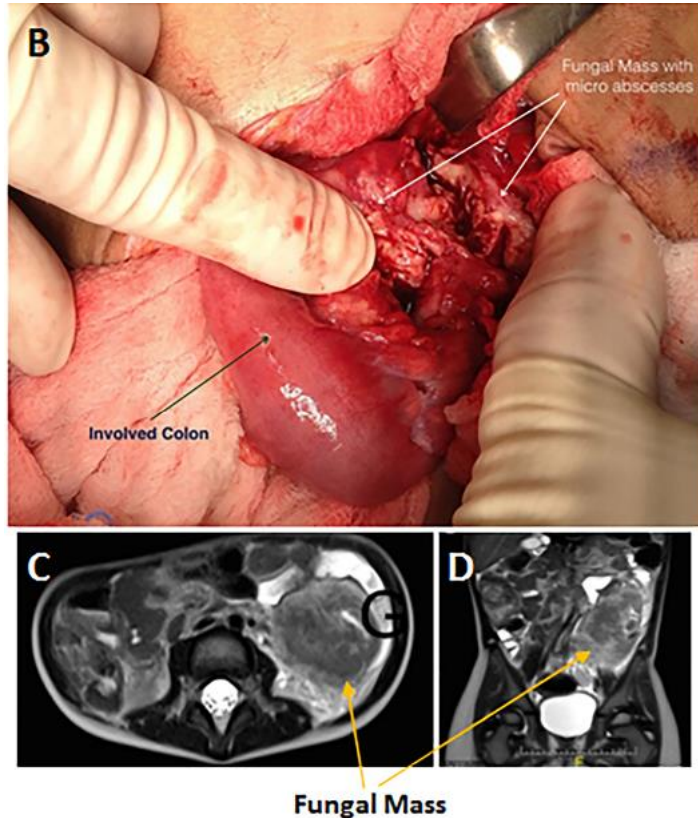
- Begins as soft tissue swelling that progress to subcutaneous nodule formation over nasal and paranasal structures
- Rhinoscopic examination may show an enlarged inferior turbinate, which is usually the initial nidus of infection
- ***Conidiobolus coronatus*** is the most frequently recovered organism
- Intracranial extension result in a rhino-orbitocerebral infection
- Extremely rare cases spread to the mediastinum, pericardium, lung, heart and thyroid

Chronic Subcutaneous Entomophthoramycosis

- Caused by *Basidiobolus* spp.
- Lesions usually are on limbs, buttocks, back and thorax
- The nodules of basidiobolomycosis cause erythema and warmth of the overlying skin due to the underlying inflammation
- Can cause obstruction of the underlying lymphatics leading to lymphedema and enlargement of the affected structure also hyperpigmentation and/or ulceration of the overlying skin



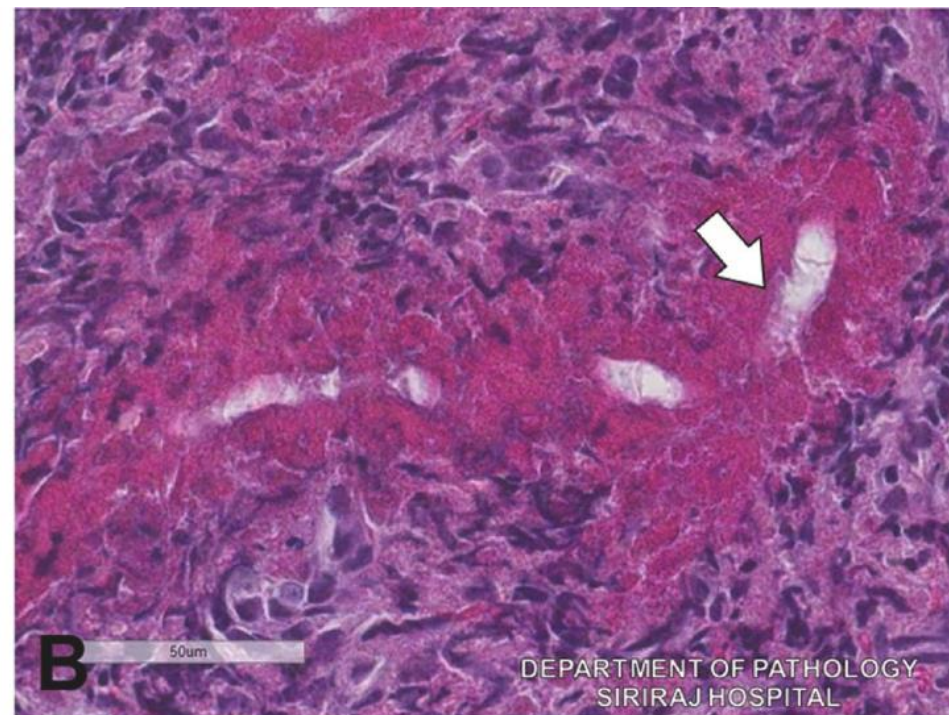
Gastrointestinal Basidiobolomycosis



- Extremely rare condition resulting from *B.ranarum* infection of gastrointestinal tract
- Clinical features include fever, abdominal pain, palpable abdominal mass, leukocytosis and eosinophilia
- Usually mimics other conditions such as intestinal lymphoma and inflammatory bowel disease, for example, Crohn's disease or ulcerative colitis

Diagnosis

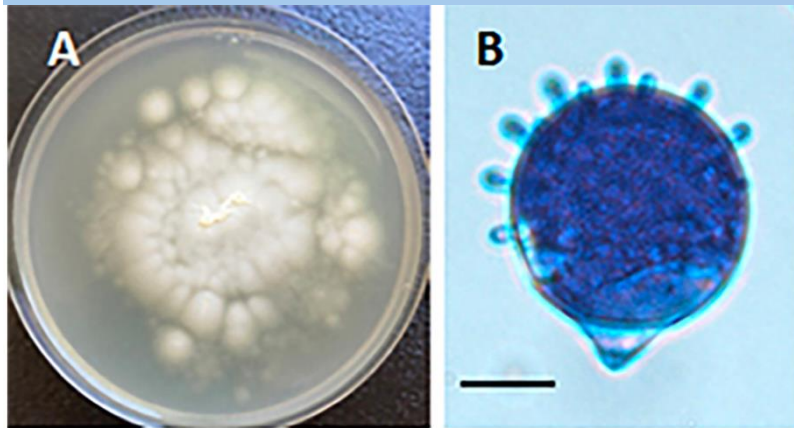
- Diagnosis is achieved by suspicion of the clinical course and by specific investigation of tissue biopsy and culture
- Histopathology shows typical findings including the **Splendore-Hoepli phenomenon**
- **Culture of infected tissue remains the definitive method of laboratory diagnosis but may have lower yield because hyphae are easily damaged during tissue processing**
- Molecular methods may be used to identify organisms in tissues when cultures are negative



Splendore-Hoepli phenomenon characterized by fungal hyphae elements surrounded by eosinophils

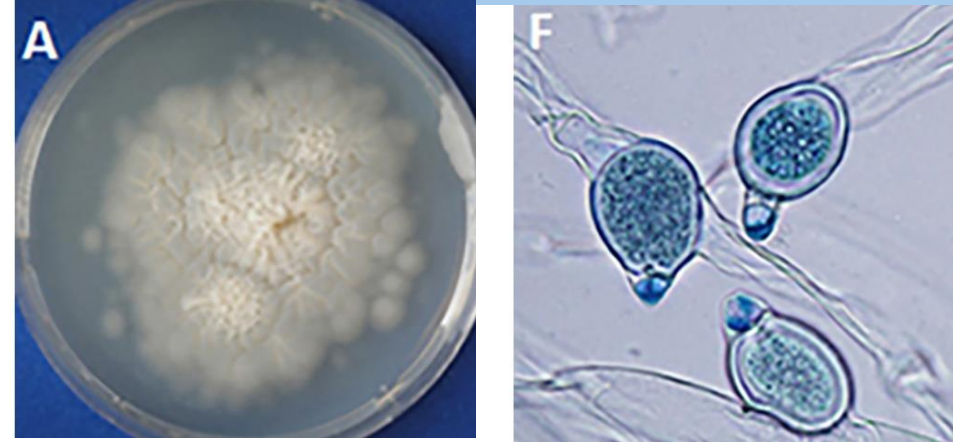
Fungus Culture

Conidiobolus coronatus



- Colony: glabrous or waxy, yellow-white, flat and wrinkled
- Microscopic: spores may have hairlike, villous projections

Basidiobolus ranarum



- Colony: glabrous or waxy, gray to beige, flat and wrinkled
- Microscopic: club shaped spores with knoblike or beaklike appendage

Treatment

Medical treatment

- Antifungal reported usage include potassium iodine (KI), co-trimoxazole, amphotericin B, ketoconazole, itraconazole, fluconazole, miconazole, voriconazole, terbinafine, and 5-fluorocytosin
- The duration of treatment ranged from several months to more than a year
- The obvious clinical response can be seen within several weeks after medication use

Surgical treatment

- May be useful in early disease of nasofacial conidiobolomycosis where cosmetic injury may be minimized but usually not feasible in advanced stages without serious cosmetic damage
- On the other hand, resection of lesions of gastrointestinal basidiobolomycosis is important in the management of this form of entomophthoromycosis

Summary

- Although rare, entomophthoramycosis should be included in the differential diagnosis of chronic subcutaneous infections, especially in immunocompetent individuals from endemic regions
- Early clinical suspicion supported by histopathology and fungal culture is crucial for timely diagnosis
- Treatment involves prolonged antifungal therapy along with surgical debridement when necessary

Progression



one month after combination treatment



5 weeks after combination treatment



Thank you

